Enhancing LLM Preference-Following Evaluation Through Realistic Preferences and Multi-Valid-Option Preference Assessment

Project Report

2. Abstract

Large Language Models (LLMs) are increasingly used in personalized applications where understanding and following user preferences is important. The PREFEVAL benchmark evaluates this capability using explicit and implicit preference-based tasks. However, its preference representations can be relatively synthetic, and its implicit multiple-choice evaluation generally assumes a single correct option, which may not fully represent realistic preference-based decision making.

This project extends PREFEVAL through two enhancements: naturalized preference representations and multi-valid-option preference assessment. Existing preferences are converted into more natural conversational forms while preserving their meaning, and enhanced MCQs allow one, two, three, or four valid options based on the underlying preference. Four experimental conditions compare original and naturalized preferences across original and enhanced MCQs. Single-answer accuracy is used for the original benchmark, while exact set match, precision, recall, and F1-score are used for the enhanced evaluation.

3. Introduction

LLMs have demonstrated strong language understanding and generation capabilities, but their effectiveness in personalized applications also depends on their ability to recognize and follow individual user preferences. PREFEVAL provides a systematic framework for evaluating this capability through explicit and implicit preference-following tasks. However, evaluating preference following through synthetic preference representations and single-correct-option questions can limit how realistically these capabilities are measured.

This project builds upon PREFEVAL by introducing more realistic preference representations and a multi-valid-option evaluation framework. Naturalized preferences represent the same underlying user preference in a conversational form, while enhanced MCQs allow multiple options to be valid when they are consistent with the preference. The proposed framework is evaluated against the original PREFEVAL baseline to study how these changes affect LLM preference-following performance.

4. Related Work / Literature Review

Recent work on Large Language Models has increasingly focused on their ability to understand and follow user preferences for personalized interactions. The PREFEVAL benchmark provides a structured approach for evaluating preference following through explicit and implicit preference tasks. Its evaluation framework measures whether an LLM can correctly identify and follow preferences when answering questions.

The baseline evaluation primarily uses predefined preference representations and MCQs with a single clearly aligned option. This project builds upon PREFEVAL by addressing these limitations through more natural preference representations and multi-valid-option MCQs, where multiple choices may satisfy the same preference. These extensions aim to make preference-following evaluation more representative of realistic situations while retaining the original PREFEVAL setup as a baseline for comparison.

5. Proposed Methodology

The proposed methodology extends the original PREFEVAL benchmark while retaining its evaluation framework as the baseline. The original preference data is first preprocessed and aligned with the corresponding MCQs. The preferences are then converted into more natural, conversational representations while preserving their original meaning. The enhanced data is checked for consistency, duplicates, and correct preference-question alignment.

The second enhancement introduces multi-valid-option MCQs, where 1, 2, 3, or 4 options can be valid according to the given preference. The evaluation compares four conditions: Original Preference with Original MCQ, Naturalized Preference with Original MCQ, Original Preference with Enhanced MCQ, and Naturalized Preference with Enhanced MCQ. Original MCQs use single-answer evaluation, while enhanced MCQs use set-based evaluation to measure exact match, precision, recall, and F1-score.

5.1 Dataset Preparation

The original PREFEVAL collection contains 1000 MCQ examples. After alignment with the naturalized and enhanced data and removal of examples that could not be consistently aligned across the required sources, the final evaluation set contains 997 examples. The aligned records retain the original preference, naturalized preference, original options, enhanced options, and corresponding valid-option information.

5.2 Naturalized Preferences

The naturalization step changes the presentation of a preference into a more conversational form without changing the intended preference. This provides a way to test whether preference following remains stable when the same information is expressed more naturally rather than in a predefined representation.

5.3 Multi-Valid-Option MCQs

Unlike the original MCQs, the enhanced questions do not require exactly one correct option. Depending on the preference, an enhanced question may contain one, two, three, or all four valid options. The final dataset contains 624 examples with one valid option, 143 with two, 78 with three, and 152 with four valid options. This makes the enhanced evaluation a set-selection task rather than a simple single-label classification task.

6. Experimental Setup

The experiments use the original PREFEVAL data together with the naturalized preferences and enhanced multi-valid-option MCQs. A total of 997 aligned examples are evaluated under the four experimental conditions. The model is given the preference, question, and answer options and is instructed to return only the required option letter or option letters.

For fair comparison, MCQ options are randomly shuffled using a fixed seed so that the correct answer is not associated with a fixed answer position. Original MCQs are evaluated using single-answer exact accuracy. Enhanced MCQs are evaluated as predicted sets against the ground-truth valid-option sets using exact set match, precision, recall, and F1-score. Raw model responses are preserved for later analysis.

6.1 Model and Evaluation Procedure

The evaluation was performed using the openai/gpt-oss-20b model through the Groq API. The same evaluation procedure was applied across the four conditions. For original MCQs, the model was

instructed to select one option. For enhanced MCQs, it was instructed to select all options that were valid according to the preference, with the possibility of 1, 2, 3, or 4 valid choices. The evaluation was run on all 997 examples and the results were stored for metric calculation.

7. Results and Analysis

The evaluation was conducted on 997 examples across four experimental settings. Original Preference with Original MCQ achieved 95.79% accuracy, while Naturalized Preference with Original MCQ achieved 96.09% accuracy. For the enhanced multi-valid-option evaluation, Original Preference with Enhanced MCQ achieved 81.95% exact set match, with 93.02% precision, 96.33% recall, and 92.90% F1-score. Naturalized Preference with Enhanced MCQ performed slightly better, achieving 82.95% exact set match, 93.01% precision, 96.88% recall, and 93.04% F1-score.

Performance varied according to the number of valid options. For Naturalized Preference with Enhanced MCQ, exact set-match accuracy was 85.58% for one valid option, 76.22% for two, 73.08% for three, and 83.55% for four valid options. For Original Preference with Enhanced MCQ, the corresponding values were 86.06%, 74.83%, 71.79%, and 76.97%. The results indicate that questions requiring the model to identify several acceptable choices are generally more difficult than the original single-answer setting.

8. Ablation Study

The ablation study examines the contribution of naturalized preference representation while keeping the MCQ type fixed. In the original MCQ setting, changing from the original preference to the naturalized preference increased accuracy from 95.79% to 96.09%. In the enhanced setting, exact set-match accuracy increased from 81.95% to 82.95%, while F1-score increased from 92.90% to 93.04%.

The improvement is relatively small, but it shows that naturalizing the preference did not reduce performance. The larger change in the overall evaluation comes from the move to multi-valid-option questions, which changes the task from selecting one answer to identifying a complete set of preference-consistent answers.

9. Discussion

The results show that the proposed framework gives a more flexible way to measure preference following. With 997 examples, the original single-answer setting produced high accuracy for both preference representations: 95.79% with the original preference and 96.09% with the naturalized preference. In the enhanced setting, exact set match decreased to 81.95% for the original preference and 82.95% for the naturalized preference. However, the enhanced setting still produced high precision and recall, with F1-scores of 92.90% and 93.04%, respectively. This difference between exact set match and the other metrics is useful because a model can identify many preference-consistent choices without necessarily producing the complete correct set.

The valid-option breakdown gives a clearer picture of this behavior. With one valid option, exact match was 86.06% for original preferences and 85.58% for naturalized preferences. With two valid options, the scores were 74.83% and 76.22%, and with three valid options they were 71.79% and 73.08%. For four valid options, the score increased to 76.97% for original preferences and 83.55% for naturalized preferences. The corresponding F1-score for the four-option case increased from 93.90% to 95.34%. Overall, the results suggest that the main challenge is not simply identifying a preference-consistent option, but identifying the full set of valid options.

10. Limitations

The evaluation is based on 997 aligned examples derived from the original PREFEVAL benchmark, so the enhanced benchmark remains dependent on the coverage and characteristics of the original dataset. Naturalized preferences may not capture every way a real user could express the same preference. In addition, deciding whether an option is genuinely consistent with a preference can require judgment when multiple choices are semantically similar.

The experiments also use one model configuration and one fixed evaluation procedure, so the results should not be treated as representative of all LLMs. The enhanced evaluation uses exact set match, which requires the model to identify the complete set of valid options. Precision, recall, and F1-score are therefore also reported to show partial performance. Future work can expand the dataset, evaluate more models, and use broader human validation for both naturalized preferences and multi-valid-option questions.

11. Conclusion and Future Work

This project extended the PREFEVAL benchmark in two ways: by representing preferences in a more natural conversational form and by introducing MCQs in which one, two, three, or four options may be valid. The evaluation retained the original PREFEVAL setup as a baseline and compared original and naturalized preferences across both single-answer and multi-valid-option settings. On 997 examples, naturalization produced a small improvement in both original and enhanced evaluations, while the multi-valid-option setting provided a more flexible and demanding measure of preference following.

Future work can evaluate a wider range of LLMs and model configurations, increase the diversity and scale of the preference data, and perform broader human validation of the enhanced questions. More complex preference scenarios and evaluation methods that measure partial correctness could also be explored.

12. References

1. PREFEVAL, Do LLMs Recognize Your Preferences? Evaluating Personalized Preference Following in LLMs.

2. PREFEVAL benchmark dataset and evaluation framework used as the baseline for this project.

3. Groq API documentation and model-serving interface used for the experimental evaluation.

4. Python Software Foundation, Python Documentation.

13. Appendix — Evaluation Prompts

Original MCQ evaluation prompt: Select the ONE option that best matches the preference. Return ONLY one option letter. Do not provide an explanation. Example: B.

Enhanced MCQ evaluation prompt: Select ALL options that are valid according to the preference. There may be 1, 2, 3, or 4 correct options. Return ONLY the option letters separated by commas. Do not provide an explanation. Example: A,C,D.

For original MCQs, the response was evaluated as a single predicted option. For enhanced MCQs, the predicted option set was compared with the ground-truth valid-option set using exact set match, precision, recall, and F1-score.

Summary of Main Results

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Original MCQ Evaluation

Condition	Accuracy / Exact Match
Original Preference + Original MCQ	95.79%
Naturalized Preference + Original MCQ	96.09%

Enhanced Multi-Valid-Option MCQ Evaluation

Condition	Exact Set Match	Precision	Recall	F1
Original Preference + Enhanced MCQ	81.95%	93.02%	96.33%	92.90%
Naturalized Preference + Enhanced MCQ	82.95%	93.01%	96.88%	93.04%

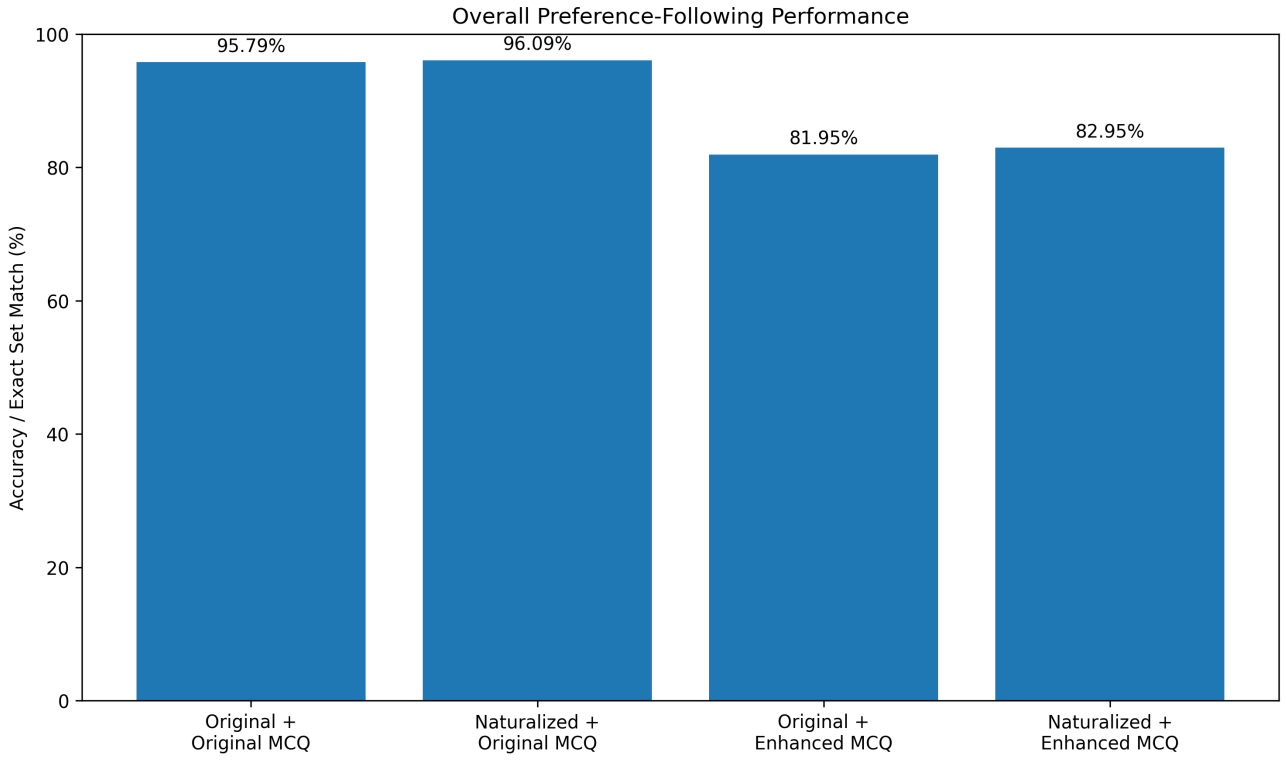


Figure: Overall preference-following performance across the four experimental conditions.